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Project Title:	Telomeric proteins in the radiation/DNA damage response		
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Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
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COI Name (Institution):	Liber, Howard (Colorado State University)		
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Task Description:	This research addresses Radiation Health Bioastronautics Roadmap (BR) Carcinogenesis Questions 28c, d and j, and is aimed at advancing current knowledge and mechanistic understanding of the DNA damage processing that underlies radiation effects in general, and space radiation in particular. The goal of the studies proposed here is to provide information that will aid identification of susceptible individuals, ultimately improving risk prediction for radiation carcinogenesis in astronauts exposed to high energy charged particles in space. Summary: Appropriate cellular response to natural chromosomal termini (telomeres) and to exogenously created double-strand breaks (DSBs) is critical to maintaining genomic stability. Incorrect handling of DSB ends leads to mutation, chromosomal rearrangements, instability and carcinogenesis. Altered levels of telomeric proteins have been reported in some tumors. The interplay between telomeres and DSBs may be better understood not only by viewing the striking differences between them, but perhaps even more importantly by viewing the obvious similarity-they are both DNA ends, the very substrate both telomere and damage response/repair proteins specifically recognize and bind. We have been investigating the role DNA repair proteins play at telomeric ends. Recent and intriguing studies suggesting that the classic telomere-binding protein TRF2 is also found (rapidly and transiently) at sites of photo-induced DNA DSBs, has turned the question around-what are telomere proteins doing at DSB ends? Here, we propose to address this fascinating question by investigating the role(s) telomeric proteins play in ionizing radiation (IR)-induced DNA damage response/processing/repair. Our approach assesses mutation frequency and spectra at the molecular level, as well as extends the analyses to the cytogenetic level. Our preliminary studies lend support to the view that telomeric proteins aren't just for telomeres anymore. Increasing our understanding of previously unrecognized relationships between the radiation/DNA damage response and telomere function will facilitate predicting the carcinogenic risk of exposure to radiation in general, and to astronauts exposed to HZE particles in particular. Ultimately, we foresee utilizing telomere instability/dysfunction as a useful biomarker of underlying repair deficiency in order to identify individual susceptibility to radiation carcinogenesis.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our investigation of interdependencies between DNA repair and telomeres will ultimately lead to identification and characterization of previously unappreciated telomere dysfunction, which in turn will provide a useful indication of an underlying repair defect. Therefore, telomere instability may ultimately serve as a sensitive biomarker, identifying individual susceptibility to radiation carcinogenesis in general and individual susceptibility to HZE exposure in particular.
Task Progress:	Our overall aim has been to study the consequences of depleting expression of critical telomere proteins on the DNA damage response induced by ionizing radiation. Background and Hypothesis: Appropriate cellular response to natural chromosomal termini (telomeres) and to exogenously created double-strand breaks (DSBs) is critical to maintaining genomic stability. Incorrect handling of DSB ends leads to mutations, chromosomal rearrangements, instability, and carcinogenesis, and altered levels of telomeric proteins have been reported in some tumors. The intriguing interplay between telomeres and DSBs may be better understood not only by viewing the striking differences between them, but perhaps even more importantly by viewing the obvious similarity – they are both DNA ends, the very substrate both telomere and damage response/repair proteins specifically recognize. This leads to the hypothesis that telomeric proteins aren't just for telomeres anymore. For example, TRF1 and TRF2 have multiple interacting binding partners, including some that are telomere specific, like Protection of Telomeres (POT)1 and tankyrase1, and some involved in DNA repair (e.g., ATM, MRN, PARP2, WRN, BLM). Telomerase is also known to interact with many telomeric, as well as non-telomeric proteins (including DNA repair proteins), and extra-curricular/pro-tumorigenic activities of telomerase independent of its classical role in telomere length maintenance, including enhanced repair, were proposed. Methods: Our approach has been to knock down expression of critical telomeric proteins in human cell lines utilizing small interfering RNA (siRNA) strategies. DNA damage was induced via exposure to ionizing radiation (low LET gamma-rays or HZE 56Fe ions). We employed a powerful combination of mutational and molecular cytogenetic analyses to examine the phenotypic and genotypic consequences of telomeric deficiencies on DNA repair endpoints. Summary of Results: We initially investigated the role DNA repair proteins play at telomeric ends. Studies suggesting that the "classic" telomere-binding protein TRF2 is also found (rapidly and transiently) at sites of photo-induced DNA DSBs, turned the question around – what are telomere proteins doing at DSB ends? We have shown that ionizing radiation (IR)-induced DSBs are insufficient to initiate TRF2 recruitment to damage sites. We further addressed this fascinating question by investigating the role(s) telomeric proteins play in IR-induced DNA damage response/processing/ repair. Depletion of either TRF2 or POT1 resulted in significantly increased spontaneous and IR-induced mutation at the heterozygous thymidine kinase (tk) locus in human lymphoblastoid cells. Both TRF2 and POT1 knockdowns behaved similarly; each led to elevated mutation and telomere dysfunction. We also investigated roles of the telomere associated Poly(ADP-ribose) Polymerase (PARP), tankyrase 1, in both telomere function and the DNA damage response following IR exposure. Tankyrase 1 knockdown significantly elevated recombination specifically within telomeres; i.e., telomere sister chromatid exchange (T-SCE), a phenotype with the potential of accelerating cellular senescence, and therefore aging. Most surprising however, was that depletion of tankyrase 1 resulted in concomitant and rapid reduction of the non-homologous end-joining (NHEJ) protein DNA-PKcs, while Ku86 and ATM protein levels remained unchanged; DNAPKcs mRNA levels were also unaffected (8). We found that the requirement of tankyrase 1 for DNA-PKcs protein stability reflected the necessity of its PARP enzymatic activity. We also demonstrated that depletion of tankyrase 1 resulted in proteasome-mediated DNA-PKcs degradation, explaining the associated defective damage response observed; i.e., increased sensitivity to IR-induced cell killing, mutagenesis, chromosome aberration, and telomere fusion. Our results suggested that tankyrase 1 mediated PARsylation may represent an important mechanism for regulating DNA-PKcs kinase activity, prompting us to consider whether pADPr modification influenced DNA-PK function in DNA repair. Investigation into the role of tankyrase dependent PARsylation of DNA-PKcs in NHEJ revealed that tankyrase PARP catalytic activity facilitated recruitment to and function of DNA-PKcs at DSB sites, providing the first insights into the relevance of regulation of DNA-PKcs by pADPr modification.

More recently, we evaluated telomerase activity following IR exposure to determine whether differences in radiation quality differentially influences telomerase activity, an important consideration for carcinogenesis with particular implications for stem cell compartments. We tracked telomerase activity and telomere length in a panel of cancer and non-cancer cell lines following both acute and low dose rate (LDR) exposures to gamma-rays. We observed elevations of telomerase activity in cancer, but not non-cancer, cell lines following acute exposures to IR. Further, telomere length was significantly reduced in both cancer and non-cancer cells postacute IR exposure. Taken together, these studies suggest telomerase activity is playing a role in accelerated tumor repopulation following radiation therapy and that the associated telomere loss may be contributing to genomic instability.

As IR induced enrichment of cancer stem cells (CSC) in established cancer cell lines was recently suggested to play a role in accelerated tumor repopulation following radiation therapy, we investigated a potential role for telomerase in the IR induced enrichment of CSCs. Consistent with previous reports, we detected a significant enrichment of breast CSCs in MCF-7 mammary carcinoma cells at 5 days post exposure, and demonstrated significant enrichment of CSCs in the non-tumorigenic MCF-10a, WTK1, and LCL15044 cell lines. Further, small molecule inhibition of telomerase activity effectively blocked CSC enrichment in both MCF-7 and MCF-10a cells. Together, these results suggest that telomerase is a significant player in the IR induced enrichment of CSCs in both cancer and non-cancer mammary epithelial cells, independent of its role in telomere length maintenance, supporting non-canonical roles of telomerase in the IR response.

Our studies continue to support the functional significance of telomeres, telomerase, and IRinduced DNA damage response interdependency, as well as the view that telomeres and DNA repair, aging and carcinogenesis, are intimately connected. Therefore, individual susceptibility factors involving both telomere function and DNA repair capacity must be considered when estimating cancer risk following ionizing radiation exposures, particularly high LET radiations. Therefore, these studies have enlightened models of cancer processes/risk in order to reduce uncertainty in radiation quality effects from galactic cosmic radiation exposure. They have also enlightened models of cancer processes/risk in order to reduce uncertainty in individual radiation sensitivity (genetic factors) from such exposure.

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